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Supersymmetric and Shape-Invariant Generalization for Non-resonant Jaynes-Cummings Systems

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Abstract

A class of shape-invariant bound-state problems which represent transitions in a two-level system introduced earlier are generalized to include arbitrary energy splittings between the two levels. We show that the coupled-channel Hamiltonians obtained correspond to the generalization of the non-resonant Jaynes-Cummings Hamiltonian, widely used in quantized theories of laser. In this general context, we determine the eigenstates, eigenvalues, the time evolution matrix, and the population inversion matrix factor.

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I INTRODUCTION

The integrability condition called shape-invariance originates in supersymmetric quantum mechanics [**?**, **?**]. The separable positive-definite Hamiltonian $\hat{H}_1 = \hat{A}^\dagger \hat{A}$ is called shape-invariant if the condition

$$\hat{A}(a_1)\hat{A}^\dagger(a_1) = \hat{A}^\dagger(a_2)\hat{A}(a_2) + R(a_2) \tag{I.1}$$

is satisfied [3]. In this equation a_1 and a_2 represent parameters of the Hamiltonian. The parameter a_2 is a function of a_1 and the remainder $R(a_1)$ is independent of the dynamical variables such as position and momentum. Even though not all exactly-solvable problems are shape-invariant [4], shape invariance, especially in its algebraic formulation [**?**, **?**], has prove, to be a powerful technique to study exactly-solvable systems.

II THE GENERALIZED NON-RESONANT JAYNES-CUMMINGS HAMILTONIAN

The Resonant Limit

The Standard Jaynes-Cummings Limit

III THE TIME EVOLUTION OF THE SYSTEM

To study the time-dependent Schrödinger equation for a Jaynes-Cummings system in non-resonant situation

$$i\hbar\frac{\partial}{\partial t}|\Psi(t)\rangle = (\hat{\mathbf{H}}_o + \hat{\mathbf{H}}_{int})|\Psi(t)\rangle \tag{III.1}$$

we can write the wavefunction as

$$|\Psi(t)\rangle = exp(-i\hat{\mathbf{H}}_o t/\hbar)|\Psi_i(t)\rangle, \tag{III.2}$$

and, by substituting this into Schrödinger equation and taking into account the commutation property between $\hat{\mathbf{H}}_o$ et $\hat{\mathbf{H}}_{int}$, we obtain

$$i\hbar\frac{\partial}{\partial t}|\Psi_i(t)\rangle = \hat{\mathbf{H}}_{int}|\Psi_i(t)\rangle \tag{III.3}$$

We introduce the evolution matrix $\hat{\mathbf{U}}_i(t,0)$:

$$|\Psi_i(t)\rangle = \hat{\mathbf{U}}_i(t,0)|\Psi_i(0)\rangle \tag{III.4}$$

which satisfies the equation

$$i\hbar\frac{\partial}{\partial t}\hat{\mathbf{U}}_i(t,0) = \hat{\mathbf{H}}_{int}\hat{\mathbf{U}}_i(t,0), \tag{III.5}$$

that is, in matrix form, written as

$$i\hbar\begin{bmatrix} \hat{U}'_{11} & \hat{U}'_{12} \\ \hat{U}'_{21} & \hat{U}'_{22} \end{bmatrix} = \alpha\begin{bmatrix} \beta & \hat{T}\hat{B}_- \\ \hat{B}_+\hat{T}^\dagger & -\beta \end{bmatrix}\begin{bmatrix} \hat{U}_{11} & \hat{U}_{12} \\ \hat{U}_{21} & \hat{U}_{22} \end{bmatrix} \tag{III.6}$$

where the primes denote the time derivative. One fast way to diagonalize the evolution matrix differential equation is by differentiating Eq. (3.5) with respect to time. We find

$$i\hbar\frac{\partial^2}{\partial t^2}\hat{\mathbf{U}}_i(t,0) = \hat{\mathbf{H}}_{int}\frac{\partial}{\partial t}\hat{\mathbf{U}}_i(t,0) = \frac{1}{i\hbar}\hat{\mathbf{H}}^2_{int}(t,0) \tag{III.7}$$

which can be written as

$$\begin{bmatrix} \hat{U''}_{11} & \hat{U''}_{12} \\ \hat{U''}_{21} & \hat{U''}_{22} \end{bmatrix} = - \begin{bmatrix} \omega_1 & 0 \\ 0 & \omega_2 \end{bmatrix} \begin{bmatrix} \hat{U}_{11} & \hat{U}_{12} \\ \hat{U}_{21} & \hat{U}_{22} \end{bmatrix} \tag{III.8}$$

where

$$\hbar\hat{\omega}_1 = \alpha\sqrt{\hat{T}\hat{B}_-\hat{B}_+ + \beta^2} = \sqrt{\hbar\Omega\hat{H}_2 + (\hbar\Delta)^2} \tag{III.9}$$

$$\hbar\hat{\omega}_2 = \alpha\sqrt{\hat{B}_-\hat{B}_+ + \beta^2} = \sqrt{\hbar\Omega\hat{H}_1 + (\hbar\Delta)^2} \tag{III.10}$$

Since by initial conditions $\hat{\mathbf{U}}_i(0,0) = \hat{I}$, , then we can write the solution of the evolution matrix differential equation (3.7) as

$$\hat{\mathbf{U}}_i(t,0) = \begin{bmatrix} \cos(\omega_1 t) & \sin(\omega_1 t)\hat{C} \\ \sin(\omega_2 t)\hat{D} & \cos(\omega_2 t) \end{bmatrix} \tag{III.11}$$

and the $\hat{C}$ the $\hat{D}$ operators can be determined by the unitarity conditions

$$\hat{\mathbf{U}}_i^\dagger(t,0)\hat{\mathbf{U}}_i(t,0) = \hat{\mathbf{U}}_i(t,0)\hat{\mathbf{U}}_i^\dagger(t,0) = \hat{\mathbf{I}}. \tag{III.12}$$

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